

AMITY UNIVERSITY

-----UTTAR PRADESH-----

Amity School of Engineering & Technology, Lucknow

Industry Internship

Student Name ABHISHEK PAL
Enrollment No A076119823010
Programme B.Tech. (Artificial Intelligence)
Company's Name and Address Maincrafts Technology
M Block, Noida Sector 66, Uttar Pradesh, India
226301

Industry Guide

Name Ujjwal Singh

Designation

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Project Information

1) Project Duration : (61 Days)

- a) Date of Summer Internship commencement (04/05/2026)
- a) Date of Summer Internship Completion (03/07/2026)

2) Topic

Automated Classification and Routing of IT Support Tickets using NLP

3) Project Objective

To design and develop an NLP-based system that automatically classifies IT support tickets into categories such as network issues, OS issues, application bugs, and security incidents, while extracting key entities to improve ticket routing and incident prioritization in IT service desks.

4) Methodology to be adopted

Collect and preprocess IT support ticket datasets. Perform text cleaning, tokenization, stop-word removal, and feature extraction using NLP techniques. Train machine learning/deep learning models such as TF-IDF + Logistic Regression, LSTM, or Transformer-based models for ticket classification. Implement Named Entity Recognition (NER) to extract important entities like device names, software names, and error codes. Evaluate the model using accuracy, precision, recall, and F1-score metrics. Deploy the trained model as a prototype web application/API for automated ticket routing.

5) Brief Summery of project(*to be duly certified by the industry guide*)

This project focuses on developing an intelligent IT support ticket management system using Natural Language Processing (NLP). The system automatically categorizes incoming support tickets into predefined domains such as network, operating system, application, and security issues. It also extracts key entities

from ticket descriptions to assist in prioritization and routing. By automating ticket classification and assignment, the proposed system reduces manual effort, improves response time, and enhances the efficiency of IT service desk operations.

Signature
(Student)

Signature
(Industry Guide)

Signature
(Faculty Guide)